1SAR refinement review

Update: what changed since 2026-05-05

- Same artefact, re-measured: every prior record left byte-identical
- R-free re-graded against the rule the T03 driver actually declares
- T14, T15 and T16 exercised on this artefact for the first time
- Partial re-run — T01/T05/T06/T10/T13 carried from May, not re-measured

The refinement was sound. The report was not — it is written about a different protein, and its headline R-free belongs to no model the agent produced.

What this re-run covers

A partial re-measurement — read the scope before the headline

The May review predates the tolerance-benchmark series. It graded its headline with a flat rule rather than the band the T03 driver declares. This run re-measures the R-factor claims and exercises three never-run tasks — it does not re-run everything.

Scope	Tasks	What that means
Re-measured here	T03 (two code paths), T14, T15, T16	graded against the current registry and driver rubric
Carried from May	T01, T05, T06, T10, T13	geometry, waters, disulfides, ligands, ions, data quality

Every conclusion about geometry, water placement, disulfides and ion identity in this deck is May's, not re-measured today. QDS_1sar_cdba2c07_2026-05-05.yaml remains the sheet of record for anything outside T03, T14, T15 and T16.

No-clobber was structural, not careful: outputs are date-keyed, the artefact zip was extracted read-only into a scratch directory, and all 102 pre-existing files were checksummed before and after. Every one is byte-identical.

Cost of that choice: this run issues a new dated .yaml where May updated the canonical one in place, so the current sheet is 271 lines against May's 1270. That is a new convention adopted here, not an existing one followed.

The R-free number, sharpened

The refinement agrees with the oracle; the report does not

May said

“R-free is reproducibly ~0.21, not 0.199.” Four code paths, 0.2116–0.2209, read as convergent corroboration.

Now measured

The agent's own model header records FREE R VALUE 0.2114. The oracle recomputes 0.2116 — agreement to 0.0002.

So the defect is

in the write-up, not the refinement. The report says 0.199 in seven places; the model it describes does not.

Where does 0.199 come from? No round the agent ran.

Model in the artefact	R-free
1sar_refined	0.2364
round 2	0.2293
round 3	0.2165
final (shipped)	0.2114
round 5	0.2109
round 6	0.2068
report claim	0.199

Every round model carries its R-free in its own PDB header, and where the oracle was re-run it agrees with the header. The range is 0.2068–0.2364. The reported 0.199 is below the best round the agent ever produced, by 0.008.

“Understated by ~0.012” implies a measurement disagreement. “Corresponds to no model produced” is a reporting-integrity problem, and a different thing to fix.

Graded against the rule T03 declares

rubric rule 1: R-free within ± 0.02 of the deposited value · the gap row is the agent's own criterion, not a registry band

Claim	What May actually graded	2026-09-07 grading	Verdict
R-free value, 0.199 vs 0.2116	all four May paths sat above 0.199	rule 1 band ± 0.02; measured |Δ| 0.0126	criterion passes, headline inaccurate
R-work/R-free gap < 0.05	0.0552 / 0.0526 / 0.0504 / 0.0435 (“3 of 4 fail”)	0.0552 PHENIX, 0.0515 gemmi	still fails
Round 6 better than final	Δ 0.0048 by oracle	Δ 0.0048, both models on the same instrument	reproduces exactly

What the independent code path did — and did not — settle

Re-run with mask radii matched to cctbx, gemmi 0.7.5 gives R-free 0.2136 against PHENIX's 0.2116 — Δ +0.0020, inside the ± 0.02 band. That settles the gemmi leg only. May's spread was 0.2116–0.2209 and its upper bound was Servalcat, which was not re-run and is not retracted; REFMAC5 likewise stands. The full bracket remains 0.2114–0.2209, with two legs carried from May.

Two cautions: the shift from May's gemmi number is confounded three ways (mask radii, version, and #316's work-only scale fit), not one. And +0.0020 sits BELOW the 0.005–0.015 offset the assumptions predict — rubric rule 2 names a smaller-than-expected gap as the red-flag direction.

Three tasks exercised on this artefact

T14 hydrogen placement · T15 structural classification · T16 interface quality

Task	Metric	Value	Reading
T14	flippable Asn/Gln/His scored — reduce / reduce2	14 / 18	0 conflicts
T14	H atoms added (Richardson reduce)	1386	band void
T15	three-state SS agreement — agent model	0.8802	uninterpretable
T15	same metric — deposited reference	0.8646	reproduces calibration
T16	DockQ vs deposited, AB:AB	0.9445	low information
T16	CAPRI interface quality class	High	low information
T16	buried surface area, chains A/B	437.8 Å²	informational

T15 is NOT a pass. The registry gates on secondary-structure CONTENT (DSSP H+E ≥ 0.20), not on agreement, because agreement is degenerate at the bad end — noise raises it. The driver does not compute content, so the governed gate could not be applied. 0.8802 is above every one of the 16 benchmark entries (0.679–0.850), and 1SAR is β-containing where the benchmark expects 0.68–0.72. A flag, not comfort.

T16 exercises the path; it does not certify an assembly. structural_criteria.yaml already classifies this contact as 1SAR's crystallographic dimer — a demonstration, not a biological assembly. With a 0.41 Å start-to-final Cα RMSD a high DockQ is near-guaranteed. Only the AB:AB mapping was scored; the swap is unmeasured.

T14 compares two genuinely independent builders (Richardson reduce, non-cctbx; mmtbx.reduce2, cctbx). Zero confident conflicts over the 14 shared residues. The H-count band is void here because the model carries SO4, CA and NA.

Coverage on the example structure

9 tasks measured · 16 of 17 settled · 1 uncovered

Measured — May, carried into this deck

- T01 superposition
- T05 geometry
- T06 model-vs-data
- T10 ligand/site
- T13 data quality

Measured — this run

- T03 X-ray refinement (re-measured, two code paths)
- T14 hydrogen placement
- T15 structural classification
- T16 interface quality

Not applicable

- T04, T08, T12 · T07 predicted-model processing · T17 NMR ensembles — none apply to an X-ray refinement artefact
- T09 molecular replacement — the model was already placed
- T11 loop fitting — no chain gaps (A 1–96, B 1–96)

Uncovered

- T02 per-residue comparison — no driver written; see next slide

An earlier draft of this deck listed T07 as covered in May. It never has been — the canonical eval carries zero T07 rows, and T07 is predicted-model processing.

Three findings the May run did not have

The first one is bigger than the R-free number

1 · The report is about the wrong protein

The artefact's final report calls 1SAR staphylococcal nuclease (SNase) — in its title, summary and discussion, 27 times. 1SAR is ribonuclease Sa from Streptomyces aureofaciens; the deposited TITLE says so. RNase Sa is 96 residues, SNase is 149.

Not a naming slip: the entire Ca^{2+} justification rests on it — “the known SNase metal-binding pocket near Asp33”, and citations to two genuine SNase papers offered as the reference for this structure's coordination. So “the ion decisions remain defensible” cannot be asserted, and this deck does not assert it.

2 · SO4 A 97 is written as ATOM

The deposition writes the sulfate as HETATM; the agent's model writes it as ATOM. A parser treating ATOM as polymer counts chain A as 97 residues instead of 96 — which happened with gemmi during this run. The agent also placed CA A 98 and NA B 98, both absent from the deposition.

3 · T02 has no headless PHENIX tool

phenix.structure_comparison is GUI-only in PHENIX 2.0 — run headlessly it exits 1 with no output and raises a project dialog. But T02 is blocked on that tool ONLY: ProSMART ships with the documented CCP4 install and May ran it on this structure, and gemmi is listed as a T02 oracle too. The gap is a missing driver, not missing tooling.

An earlier draft of this deck claimed ProSMART was not installed. That was wrong — the check was run without sourcing the CCP4 environment. Corrected above.

Trust invariant: passes, by omission

Worth stating plainly rather than claiming a win

The trust model forbids grading PHENIX solely with PHENIX. Since the 2026-08-13 cutover a sheet carrying a cctbx-only task must waive it explicitly or close it with an independent oracle. Sheets issued before the cutover are grandfathered and listed by name.

Sheet	Tasks	Gap status	Invariant
QDS_1sar…04-24 through 05-05	T06 (and T13 on three sheets)	open — cctbx only	grandfathered
QDS_1sar…2026-09-07	T03, T14	closed	satisfied
QDS_1sar…2026-09-07	T15, T16	non-cctbx only	satisfied

The new sheet passes because it does not grade T06 — the row that was open on every prior 1SAR sheet — since this run did not re-measure T06. Nothing that was open was closed. The sheet is 271 lines where May's was 1270, and no longer carries geometry, data quality, ions, waters or pairwise comparisons.

What did genuinely improve: mmtbx.reduce2 is now a registered tool. It was a dependency of the T14 benchmark but had never been in the catalog, so until now no T14 record could name the builder that produced its second opinion.

The harness since May

Why the same artefact grades differently now

48-round tolerance series

Every [template] threshold that mattered here replaced by a measured band with stated preconditions — including the R offset that scoped the gemmi comparison.

Negative-control track

0 of 22 false verdicts on the control set. Separately, 21 of 21 agent cases judged — a different denominator, since 6XVM was never judged.

Gate consolidation

Negative-control headlines, governed thresholds and round-count claims are now data checked by the gate, not prose the gate hopes someone re-read.

Toolchain routing

gemmi pinned at 0.7.5 and routed through toolchain.py, so a run can state which binary produced a number instead of inferring it.

Transfer, CI, licensing

Repository public under CultureBotAI with CI green — achieved by the transfer, which escaped a personal-account Actions billing lock. BSD-3-Clause code, CC-BY-4.0 docs.

Net assessment

What holds, what fails, what is still open

Holds (re-measured)

✓ Oracle R-free reproduces May to four decimals

✓ R-free criterion passes under rule 1's ± 0.02

✓ Both H builders agree on every confident flip

✓ Deposited T15 and T16 legs reproduce calibrations

✓ Round-6-vs-final delta reproduces exactly

Fails

✗ The report describes the wrong protein

✗ …so the stated basis for the Ca^{2+} call is void

✗ Reported R-free 0.199 matches no model produced

✗ The shipped final is not the best round

✗ SO4 written as ATOM, not HETATM

Open

T02 has no driver · T15 needs a secondary-structure content figure before its number means anything · the T16 swapped chain mapping is unmeasured · REFMAC5 and Servalcat R-free legs were not re-run · geometry, waters, ions and data quality are carried from May, not re-measured.

The R-free criterion actually passes under the rule this task declares. What does not pass is the write-up: a headline number that describes no model in the artefact, and a report written about a different protein than the one it refined — with the metal-site reasoning inherited from that mistake.